

FELKER AAF, VA, USAWMO: **723087**Lat: **37.133N**Lon: **76.600W**Elev: **4**StdP: **101.28**Time Zone: **-5.00 (NAE)**Period: **96-19**WBAN: **93735****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-4.6	-17.0	0.8	-5.0	-14.3	1.1	-2.7	10.1	0.6	8.8	1.5	3.3	10	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	34.2	25.7	33.0	25.3	31.8	24.7	28.1	30.6	27.0	30.2	26.2	29.7	2.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.7	23.7	28.7	26.3	21.8	28.2	25.2	20.4	27.8	90.1	30.7	84.9	30.2	81.2	29.8	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.1	5.8	-9.4	36.4	2.4	1.6	-11.2	37.5	-12.6	38.4	-14.0	39.3	-15.8	40.4		
(5)			-10.8	28.9	2.1	1.9	-12.3	30.3	-13.6	31.4	-14.8	32.4	-16.3	33.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	4.2	6.4	9.7	15.7	20.7	25.0	26.9	26.1	23.3	17.5	10.5	7.4
	DBStd	8.97	5.29	5.56	5.30	4.56	3.86	2.82	2.33	2.11	3.07	4.31	4.28	4.97
	HDD10.0	556	191	127	74	6	0	0	0	0	0	2	44	112
	HDD18.3	1815	438	334	270	104	23	1	0	0	2	68	236	340
	CDD10.0	2813	12	27	65	178	332	451	525	500	399	233	60	32
	CDD18.3	1031	0	1	3	27	97	201	267	242	151	41	2	1
	CDH23.3	8883	0	4	21	199	673	1786	2718	2178	1061	237	5	1
	CDH26.7	3197	0	0	1	45	188	666	1132	799	319	46	1	0
Wind	WSAvg	2.7	2.9	3.1	3.2	3.1	2.7	2.5	2.4	2.2	2.5	2.6	2.6	2.6
Precipitation	PrecAvg	1192	83	74	97	89	94	104	139	135	121	96	70	87
	PrecMax	1694	159	191	221	179	156	210	232	284	484	225	226	194
	PrecMin	830	29	29	5	36	19	35	43	42	7	0	3	30
	PrecStd	206	35	38	42	34	34	40	49	69	102	51	49	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.4	24.0	25.6	30.6	32.4	35.5	36.4	34.7	33.4	30.2	23.9	22.6
		MCWB	16.8	17.7	17.1	21.2	24.1	25.5	26.9	26.4	23.7	23.4	18.4	20.1
	2%	DB	17.5	20.9	23.1	27.8	30.3	33.4	34.7	33.3	31.7	28.0	21.5	20.0
		MCWB	13.8	15.1	15.7	19.2	22.9	25.1	26.2	25.6	24.1	22.1	17.7	17.8
	5%	DB	15.3	18.3	20.8	25.7	28.9	31.9	33.2	32.0	30.2	26.1	19.4	17.7
		MCWB	12.3	13.9	14.6	18.7	22.4	24.7	25.7	24.6	23.6	20.8	15.9	15.8
	10%	DB	12.8	15.8	18.5	23.6	27.2	30.5	31.9	30.7	28.6	24.2	17.7	15.4
		MCWB	10.0	11.9	14.2	18.0	21.3	23.9	25.3	24.2	22.9	20.3	15.1	13.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	18.7	19.0	23.6	25.3	28.3	30.2	28.7	27.5	25.1	20.7	21.3
		MCDB	19.2	22.2	22.8	25.3	28.5	30.3	31.5	31.5	28.3	27.6	21.5	21.5
	2%	WB	15.1	16.9	17.6	21.4	24.1	26.8	28.8	27.4	26.0	23.7	18.7	17.8
		MCDB	17.2	19.8	20.7	25.4	28.5	30.1	30.8	30.4	28.5	26.3	20.3	19.3
	5%	WB	12.6	15.0	16.4	19.9	23.3	25.8	27.5	26.5	25.2	22.6	17.2	16.1
		MCDB	14.7	17.8	19.8	24.4	27.3	29.8	30.9	29.8	28.0	25.1	18.8	17.7
	10%	WB	10.2	12.1	14.7	18.5	22.3	25.1	26.5	25.7	24.4	20.9	15.3	13.7
		MCDB	12.3	15.0	17.9	22.8	26.0	29.2	30.3	29.0	27.0	23.6	17.4	15.0
Mean Daily Temperature Range	5% DB	MDBR	8.7	9.8	9.8	10.1	9.3	9.0	8.9	8.5	8.3	9.2	9.2	8.6
		MCDBR	12.7	13.3	13.1	12.2	10.9	10.4	10.3	9.9	9.9	10.6	11.4	10.1
		MCWBR	10.4	9.4	7.8	6.0	5.3	4.7	4.2	4.2	4.6	5.6	7.8	9.1
	5% WB	MCDBR	11.3	13.0	10.8	10.5	9.4	9.5	9.4	8.8	8.3	8.7	9.7	9.9
		MCWBR	10.4	10.4	8.1	6.4	5.0	5.4	5.2	4.4	4.7	5.4	8.3	9.3
Clear-Sky Solar Irradiance	taub	0.305	0.318	0.346	0.392	0.434	0.487	0.516	0.511	0.409	0.354	0.322	0.309	
	taud	2.537	2.497	2.450	2.331	2.244	2.147	2.082	2.083	2.357	2.507	2.547	2.559	
	Ebn at Noon	891	920	919	889	853	805	778	772	842	866	862	862	
	Edh at Noon	77	91	105	125	138	152	161	158	114	89	76	72	
All-Sky Solar Radiation	RadAvg	2.34	3.09	4.11	5.22	5.89	6.35	6.04	5.45	4.43	3.51	2.65	2.08	
	RadStd	0.19	0.34	0.33	0.39	0.53	0.42	0.30	0.29	0.38	0.41	0.22	0.17	